The effects of mutations G350D and R710G on dynamin-related protein 1

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Abstract:

Mitochondrial fission is regulated by dynamin-related protein 1 (Drp1). The role of this protein is to form helices and constrict the mitochondrial membrane upon GTP hydrolysis, eventually leading to mitochondrial fission. Mutations in Drp1 can cause structural and biochemical changes in the protein, leading to impaired mitochondrial fission. These mutations can also cause neurodevelopmental diseases in children, although the mechanism by which the mutations progress to the diseases is unknown. This study examines the effects of mutations G350D and R710G on Drp1 through bacterial expression constructs and functional studies involving GTPase assays and negative stain electron microscopy (EM). We expect that the G350D mutation will cause an assembly defect as the mutation is located in an assembly region of the protein, and the R710G mutation may impair GTPase activity as it is located in a helix adjacent to the GTPase domain. Analyzing the effects of mutations on Drp1 assembly and function is vital to understanding the mechanism by which neurodevelopmental diseases can arise, allowing us to gain knowledge on potential treatment options. Continuing to research mitochondrial dynamics is vital to fully understanding the role that mitochondria have in our body and improving human health.

Introduction and Background:

Mitochondria are responsible for producing ATP (adenosine triphosphate) through cellular respiration. Some cells in the body have higher energy demands for mitochondrial function; therefore, it is essential to generate more ATP in these cells. Mitochondrial dynamics, dictated by the opposing processes of fusion and fission, change the morphology of the mitochondria, impacting ATP production and distribution. (1) Fusion is when two mitochondria encounter each other, merge, and become one. Mitochondrial fission is the process in which a singular mitochondria can split and develop into two or more mitochondria. In order for these cells to maintain healthy function, a balance between mitochondrial fission and fusion is crucial. (2)

Fission is regulated by dynamin-related protein 1 (Drp1), a large GTPase that forms helices and constricts the mitochondrial membrane upon GTP hydrolysis eventually leading to mitochondrial fission. (3-4) This protein is composed of four domains (Figure 1) that are vital for allowing Drp1 to constrict around the mitochondrial membrane: a GTPase domain, a GTPase effector domain (GED), a middle domain (MD), and a variable domain (VD). The role of the GTPase domain is to bind and hydrolyze GTP to initiate conformational changes throughout the rest of the protein. The MD regulates self-assembly instructions for Drp1 and forms a helical structure that interacts with the GED, called the stalk. The Bundle Signaling Element (BSE) is a region that connects the GTPase domain to the stalk. Additionally, the BSE changes shape in response to nucleotide binding to allow the nucleotide state of the GTPase domain to direct stalk reorganization of the assembly and place force on the membrane. The VD is made up of a 130 amino acid sequence which connects the ends of the MD and the GED. This domain was intrinsically disordered and has been deleted in previous studies. (5)

Mutations in the Drp1 protein and other proteins that regulate mitochondrial fission and fusion can cause neurodegenerative diseases. (6) Although it is known that mutations in the Drp1 protein cause neurodevelopmental/neurodegenerative diseases, the method by which the mutations progress to the diseases is still being researched. Furthermore, even though synthetic mutations in the Drp1 protein have helped determine the functional role of domains and patient mutations have been studied, the way in which Drp1 activity and mitochondrial fission are impaired is uncertain. In this study, mutations G350D and R710G are being introduced into bacterial expression constructs that allow the Drp1 protein to be isolated. This will allow us to study how specific mutations in Drp1 affect biochemical and structural function.

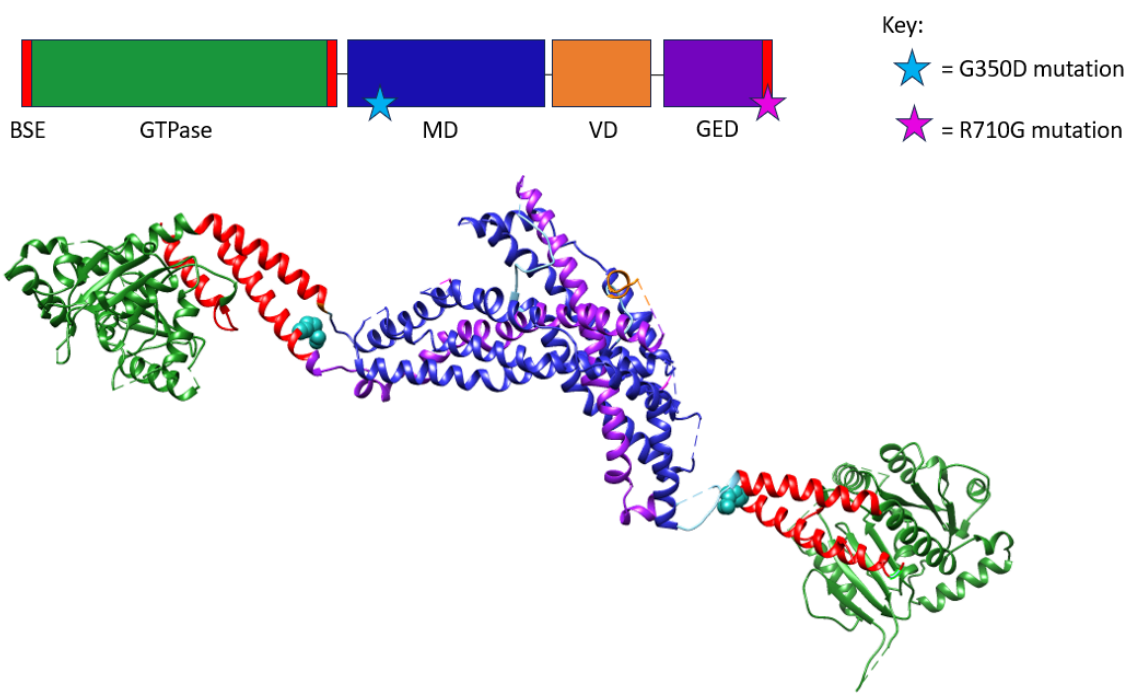

Figure 1. A picture depicting the domains of Drp1 chains A and B is displayed. The green region displays the GTPase domain. The red region displays the BSE. The blue represents the MD. The purple displays the GED. The cyan blue spheres represent the R710G mutation being characterized in this study. The magenta spheres represent the G350D mutations being characterized in this study.

To determine the effects the mutations have on the protein, we will characterize the functional differences between the mutated Drp1 and a control sample consisting of the unmutated Drp1. This will advance our understanding of the mechanisms by which the mutations progress to neurodegenerative diseases. We expect that the G350D mutation will cause an assembly defect as the mutation is located in the MD of the protein. Additionally, we expect the R710G mutation to impair GTPase activity as the mutation is located in the BSE, adjacent to the GTPase domain.

Methods:

Location:

Most assays and methods listed below were completed in the Mears lab. The negative stain EM was conducted in an EM facility at the Cleveland Center for Membrane and Structural Biology (CCMSB, https://case.edu/medicine/ccmsb/).

Bacterial Culture:

Drp1 Isoform 3 was inserted into the DH5-Alpha competent *E. coli* cell mixture using the High Efficiency Transformation Protocol. Following an established protocol, we thawed the bacterial cells on ice and added plasmid DNA (deoxyribonucleic acid) to the cells. The cells were incubated on ice for thirty minutes, heat shocked for 30 seconds at 42°C, and placed in ice for five minutes. After introducing the DNA to the bacterial cells, the streak plate procedure was used to grow the bacteria into colonies. Using this procedure, fifty µL from the DH5-Alpha competent *E. coli* cell mixture was taken and placed on an agar plate. The mixture was spread on the plate in different directions and placed in an incubator overnight at 37°C. After the colonies had grown, one colony was chosen and placed into a culture tube containing Luria broth (LB) and ampicillin and put into an incubated shaker overnight at 37°C to start the bacterial culture.

DNA purification:

DNA purification was completed using the Macherey-Nagel Plasmid DNA purification kit. We placed the culture tube of bacteria in the centrifuge at 25°C at maximum speed to harvest the bacterial cells. Following centrifugation, the supernatant containing the debris was isolated from the cell pellet. The supernatant was discarded and the majority of the liquid was removed. 250 µL of buffer A1 with RNase was added to the tube and the cell pellet was resuspended by pipetting up and down. 250 µL of buffer A2 was added to promote cell lysis. 300 µL of buffer A3 was added to initialize the effects of the previous buffer. To clarify the lysate, the tube was centrifuged for five minutes at room temperature. The plasmid column was placed into a collection tube and centrifuged for five minutes to separate the cell debris from the DNA. 700 µL of the supernatant was added to the column and centrifuged for one minute, the flowthrough was discarded and the column was placed back into a collection tube. We proceeded to wash the silica membrane using 500 µL of Buffer AW preheated to 50°C. The collection tube was centrifuged for one minute. 600 µL of Buffer A4 was added and the collection tube was centrifuged for one minute, the flowthrough was discarded. The silica membrane was dried by centrifuging for two minutes. The collection tube was discarded. To elute the DNA the plasmid column was placed in a microcentrifuge tube and 50 µL of Buffer AE was added. The solution was incubated for one minute at room temperature and centrifuged for one minute.

Mutagenesis:

Primers with mismatched base pairs to introduce G350D and R710G mutations in the bacterial expression plasmid were ordered from Integrated DNA Technologies. Using the provided specification sheet, specific volumes of water were added to suspend the primer solutions at the right concentration. A Nanodrop Spectrophotometer was used to measure plasmid and primer concentrations. A Polymerase Chain Reaction (PCR) was performed using the QuikChange Lightning Site-Directed Mutagenesis Kit. Based on the concentration results from the Nanodrop Spectrophotometer, the amount of RNase-Free Water and primer that should be added to the culture tubes were calculated using the following formula: $C_1 \times B_1 = C_2 \times B_2$. After adding the RNase-Free Water and the primers, we took two culture tubes and labeled the first one for the G350D mutation and the second one for R710G. 34 µL of RNase-Free Water, 5 µL of 10x RxN buffer, and 5 ng/µL of pCal were added to both tubes. 1.25 µL of the G350D Reverse and Forward primers were added to tube 1. Similarly, 1.25 µL of R710G Reverse and Forward primers were added to tube 2. 1 µL of dNTP (deoxynucleotide triphosphate) mix, 1.5 µL of quick solution, 1 µL of QuikChange lightning enzyme were then added to both tubes. Both PCR reaction tubes were then incubated with DpnI to degrade the template plasmid. They were then stored in the freezer until further use.

After introducing the mutations into the Drp1 Isoform 3 plasmid, we used the High Efficiency Transformation Protocol for the process of bacterial transformation. 2 µL of the PCR reaction mixture was added to the NEB 5-alpha Competent *E. coli*. They were incubated in ice for thirty minutes. The cells were heat shocked at 42°C for thirty seconds and set the cells on ice for five minutes. 950 µL of SOC Outgrowth Medium was added to the culture tubes and put in an incubated shaker at 37°C. The culture tubes were placed in a centrifuge for one minute at a speed of 10,000. We pipetted the cell mixture up and down to remove the cell pellet from the bottom of the tube. 50 µL of the cell mixture was plated and spread on a plate containing LB and ampicillin. They were left overnight in the incubator at 37°C.

Mutagenesis was performed for a second time with newly sequenced primers ordered from Integrated DNA Technologies. The second time, staggered primers were included with the normal G350D reverse and forward primers. The same steps listed above were followed for a second time.

Negative Stain Electron Microscopy:

Samples containing 2 µM Drp1 in the Assembly Buffer were prepared. On a piece of parafilm, 5 µL of WT (wild type) Drp1 protein isoform 3 was added along with two drops of 2% uranyl acetate. Carbon-coated mesh grids were placed, carbon side down, onto the drop of the protein sample. The grid remained on the drop for sixty seconds and was blotted on a piece of filter paper to remove the excess sample. The grid was washed by being placed on the first drop of 2% uranyl acetate and immediately blotted on the filter paper. The wash was repeated. The grid was then placed on the second 2% uranyl acetate for sixty seconds. The grid was blotted on filter paper to remove excess stains. An electron microscope was then used to image the grid. Grids were also created for WT Drp1 protein isoform 3 in the presence of the nucleotide analog GMPPCP. An electron microscope was used to image Drp1 in the presence of nucleotides (7).

GTPase Assay:

GTP and Drp1 were thawed on ice. The phosphate standard was prepared in an assembly buffer using Potassium Phosphate Monobasic diluted to 100, 80, 60, 40, 20, 10, 5, and 0 µm. 20 µL of the appropriate standard was added to the corresponding well. 3×GTP/MgCl2 solution in Assembly Buffer was prepared and stored in ice until use. 2.4x solution of Drp1 in Assembly Buffer was prepared and stored on ice. We added 50 µL 2.4x Drp1 to the PCR tubes and then 30 µL of Assembly Buffer. The PCR tubes were moved to a 37°C heat block and the GTP hydrolysis reactions were started after adding 40 µL 3× GTP/MgCl2. At specific time periods, 20 μL of the reaction mixture was removed and added to a well containing 5 μL 0.5 M EDTA. 150 μL of Malachite Green Reagent was added to each well and the plate was read at $A_{650.}$ The A_{650} values for the phosphate standards were plotted to create a standard curve. This curve was then used to calculate the amount of phosphate released at each time point. (7)

Results:

Through site-directed mutagenesis, the R710G reverse and forward primers, the R710G reverse and forward staggered primers, and the G350D reverse and forward staggered primers were able to generate mutated plasmid based on the successful transformation of competent *E.coli* cells (Figures 2-5). Conversely, the G350D reverse and forward primers were unable to generate colonies. DNA will be isolated from these colonies and the mutations confirmed by Sanger sequencing performed by ATCG, Inc.

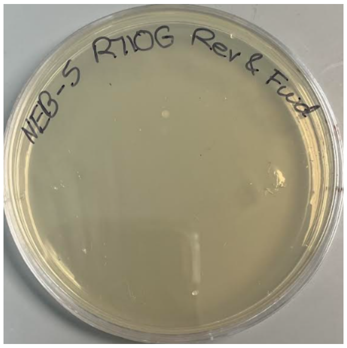

Figure 2. This picture displays a colony grown after bacterial transformation of the NEB-5 cells in the R710G reverse and forward primers.

Figure 3. This picture displays a colony grown after bacterial transformation of the NEB-5 cells in the R710G reverse and forward staggered primers.

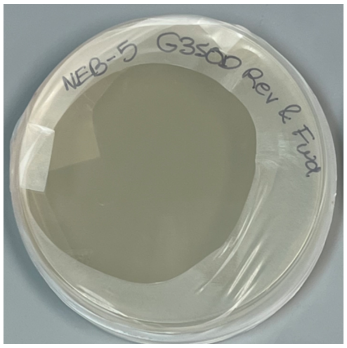

Figure 4. This picture displays no colonies grown after bacterial transformation of the NEB-5 cells in the G350D reverse and forward primers.

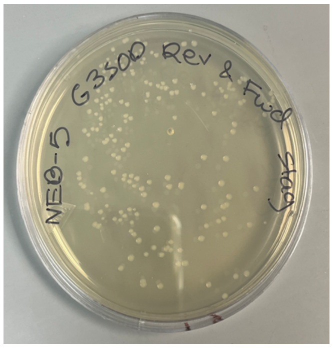

Figure 5. This picture displays colonies grown after bacterial transformation of the NEB-5 cells in the G350D reverse and forward staggered primers.

To initiate functional studies to compare with the mutant proteins, WT Drp1 was assessed using a variety of assays. By using negative stain EM, we determined that without GMP-PCP, WT Drp1 forms multimers that assume the shape of crescent particles (Figure 6). In the presence of GMP-PCP, WT Drp1 assembles into small spiral structures (Figure 7), demonstrating its ability to assemble. Additionally, a GTPase assay was performed to analyze the GTPase activity of WT Drp1 (Figure 8, see next page). From this assay, we found that WT Drp1 has an average activity of 2 μM phosphate/minute/μM protein or 2 min^{-1}, corresponding to phosphate concentration generated per minute per concentration of protein. By comparing the functional studies listed above of the mutated sequence and the WT, the effects of the G350D and R710G mutations on Drp1 structure and function can be determined.

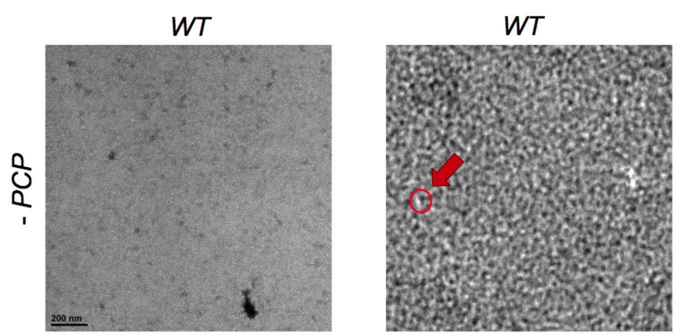

Figure 6. This picture displays a negative stain EM image of WT Drp1. In the absence of GMP-PCP, Drp1 forms small multimers that assume the shape of crescent particles (highlighted by the red circle and arrow). The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

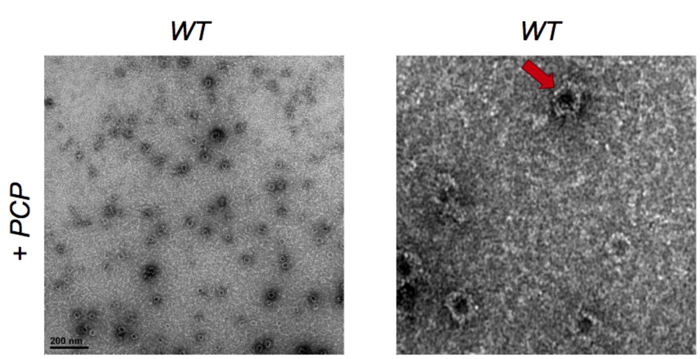

Figure 7. This picture displays a negative stain EM image of WT. In the presence of GMP-PCP, Drp1 forms larger spiral structures (highlighted with a red arrow), demonstrating its ability to assemble. The image on the right is a close-up of the same negative stain EM image of WT shown on the left.

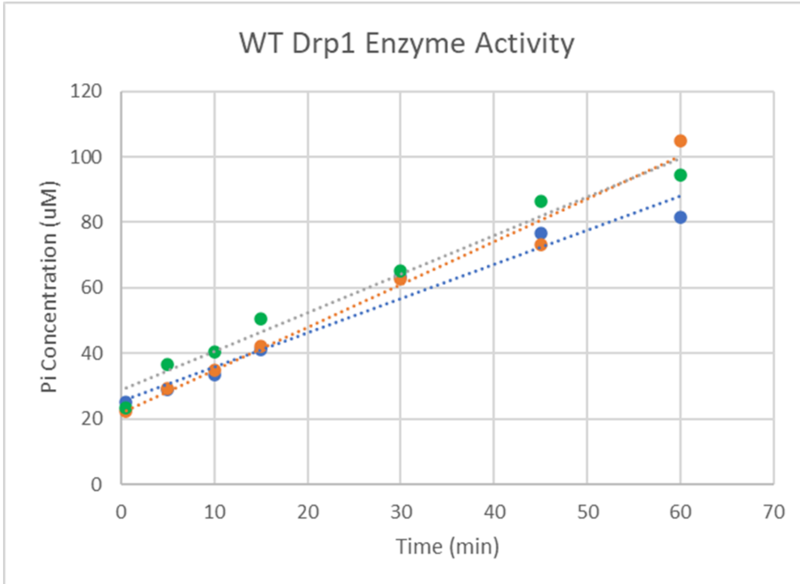

Figure 8. This picture displays WT Drp1 GTPase activity. The average slope of all three samples shows that WT Drp1 has an average activity of 2 μM phosphate/minute/μM protein or 2 min^{-1}.

Discussion:

A colony/colonies of bacteria were present in the petri-dishes with the R710G reverse and forward primers, the R710G reverse and forward staggered primers and the G350D reverse and forward staggered primers. This indicates that the primers were functional and that site-directed mutagenesis worked, allowing us to verify if the correct sequence change has been made. However, no colonies had grown in the G350D reverse and forward primers, suggesting that the primers were defective. After analyzing the images from the negative stain EM, we saw that without GMP-PCP, WT Drp1 formed multimers that assumed the shape of crescent particles. When GMP-PCP was added, the protein formed small spiral structures, showing that WT assembles in the presence of a nucleotide that impersonates the GTP-bound state that wraps around mitochondria. Additionally, a GTPase assay was performed to analyze the GTPase activity of WT Drp1. From this assay, it was concluded that WT Drp1 has an average turnover of 2 min^{-1}. This means that for each additional minute, there is approximately a 2 μM increase in phosphate per μM protein. These functional studies conducted with the WT Drp1 serve as a control group that can be compared with the mutated Drp1 sequence to analyze the effects of the G350D and R710G mutations on protein structure and function. A recent study found that the mutations A395D, R403C, and C431Y could not assemble into spirals when the nucleotide was introduced (5). These positions are located in the MD. Similarly, in our study, the G350D mutation is also located in the MD. Hence, the study mentioned above supports our hypothesis of the G350D mutation being assembly defective by enforcing the idea that mutations in the MD can affect the assembly of Drp1. Therefore, we anticipate that the G350D mutation is assembly defective as it is located in the MD and cannot form the spiral structures as shown in Figure 3. We expect the R710G mutation to impair GTPase activity, therefore we expect it to exhibit lower GTPase activity rates when compared with WT Drp1.

Implications:

Studying mutations such as G350D and R710G is crucial to understanding the mechanisms by which mitochondrial fission is inhibited and how neurodevelopmental and neurodegenerative diseases can arise, allowing us to explore potential treatment options. However, more research must be conducted to fully explore these studies at the cellular level and gain insight into potential treatments for neurological disorders. Other assays in the lab such as size-exclusion chromatography with multi-angle light scattering can be used to understand the effects of these mutations in more depth. A limitation of our study is that we used isolated protein. Bioenergetics and cell health can be assessed to complement the studies with isolated proteins. This allows us to see if interactions are affected. Additionally, these mutations can be incorporated into a cell line to observe how impaired fission would impact Drp1's function.

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References

1) Francy CA, Alvarez FJ, Zhou L, Ramachandran R, Mears JA. The mechanoenzymatic core of dynamin-related protein 1 comprises the minimal machinery required for membrane constriction. J Biol Chem [Internet]. 2015 May 1 [cited 2023 June 27];290(18):11692-703. Available from: https://science/article/pii/S0021925820570051?via%3Dihub DOI: 10.1074/jbc.M114.610881

2) Tilokani L, Nagashima S, Paupe V, Prudent J. Mitochondrial dynamics: an overview of molecular mechanisms. Essays Biochem [Internet]. 2018 Jul [cited 2023 June 6]; 62(3):341–60. Available from: http://essays.biochemistry.org/content/62/3/341 DOI: 10.1042/ebc20170104

3) Smirnova E, Griparic L, Shurland DL, van der Bliek AM. Dynamin-related protein Drp1 is required for mitochondrial division in mammalian cells. Mol Biol Cell [Internet]. 2001 Aug [cited 2023 June 30];12(8):2245–56. Available from: https:///doi/10.1091/mbc.12.8.2245 DOI: 10.1091/mbc.12.8.2245

4) Banerjee R, Kumar A, Satpati P, Nagotu S. Mimicking human Drp1 disease-causing mutations in yeast Dnm1 reveals altered mitochondrial dynamics. Mitochondrion [Internet]. 2021 Jul [cited 2023 June 27];59:283-295. Available from: https://science/article/abs/pii/S1567724921000829?via%3Dihub doi: 10.1016/j.mito.2021.06.009.

5) Bauer BL, Rochon K, Liu JC, Ramachandran R, Mears JA. Disease-associated mutations in Drp1 have fundamentally different effects on the mitochondrial fission machinery. Hum Mol Genet [Internet]. 2023 Jun 5 [cited 2023 June 8];32(12):1975-1987. Available from: https://pubmed.ncbi.nlm.nih.gov/36795043/ doi: 10.1093/hmg/ddad029.

6) Knott AB, Perkins G, Schwarzenbacher R, Bossy-Wetzel E. Mitochondrial fragmentation in neurodegeneration. Nat Rev Neurosci [Internet]. 2008 Jul [cited 2023 June 21];9(7):505-18. Available from: https://pubmed.ncbi.nlm.nih.gov/18568013/ doi: 10.1038/nrn2417

7) Clinton RW, Bauer BL, Mears JA. Affinity Purification and Functional Characterization of Dynamin-Related Protein 1. Methods Mol Biol [Internet]. 2020 Jan 1 [cited 2023 Jul 13];2159:41–53. Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8018536/